POWER BI – WEEK 10 PROJECT REPORT

HOUSE PRICE ANALYSIS DASHBOARD USING MICROSOFT POWER BI

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1. Introduction

Microsoft Power BI is a Business Intelligence tool used to analyze data and create interactive dashboards. This project focuses on analyzing house sales data using Power BI. The kc_house_data dataset is used to explore house prices, property sizes, bedrooms, bathrooms, waterfront details, and other property features.

The main aim of this project is to create an interactive dashboard with different visualizations, KPI cards, filters, and geographical analysis to understand housing market trends.

2. Objectives

- To analyze housing sales data using Power BI.
- To create an interactive house price analysis dashboard.
- To calculate important Key Performance Indicators (KPIs).
- To analyze house prices based on different property features.
- To visualize property locations using a map.
- To create charts for comparing house prices and living areas.
- To use slicers and filters for interactive data analysis.
- To create DAX measures and calculated tables.

3. Tools and Technologies Used

- Microsoft Power BI Desktop: Used to create dashboards and visualizations.
- Power Query Editor: Used for data preparation and transformation.
- DAX (Data Analysis Expressions): Used to create measures and calculations.
- kc_house_data: Housing sales dataset used for analysis.

4. Dataset Description

The kc_house_data dataset contains residential property sales information. It includes details about house prices, bedrooms, bathrooms, living area, waterfront, condition, grade, floors, and geographical location.

These attributes help to understand the relationship between property features and house prices.

5. Dashboard Creation

The project dashboard was created using multiple report pages. Different visualizations were used to represent housing data in a clear and understandable format.

6. Page 1 – House Price Analysis Dashboard

The first page contains the main dashboard title and five KPI cards.

6.1 Total Properties Sold

A KPI card was created to display the total number of properties sold. It provides an overview of the number of property sales available in the dataset.

6.2 Average House Price

This card displays the average selling price of houses. It helps understand the overall average property price.

6.3 Median Price

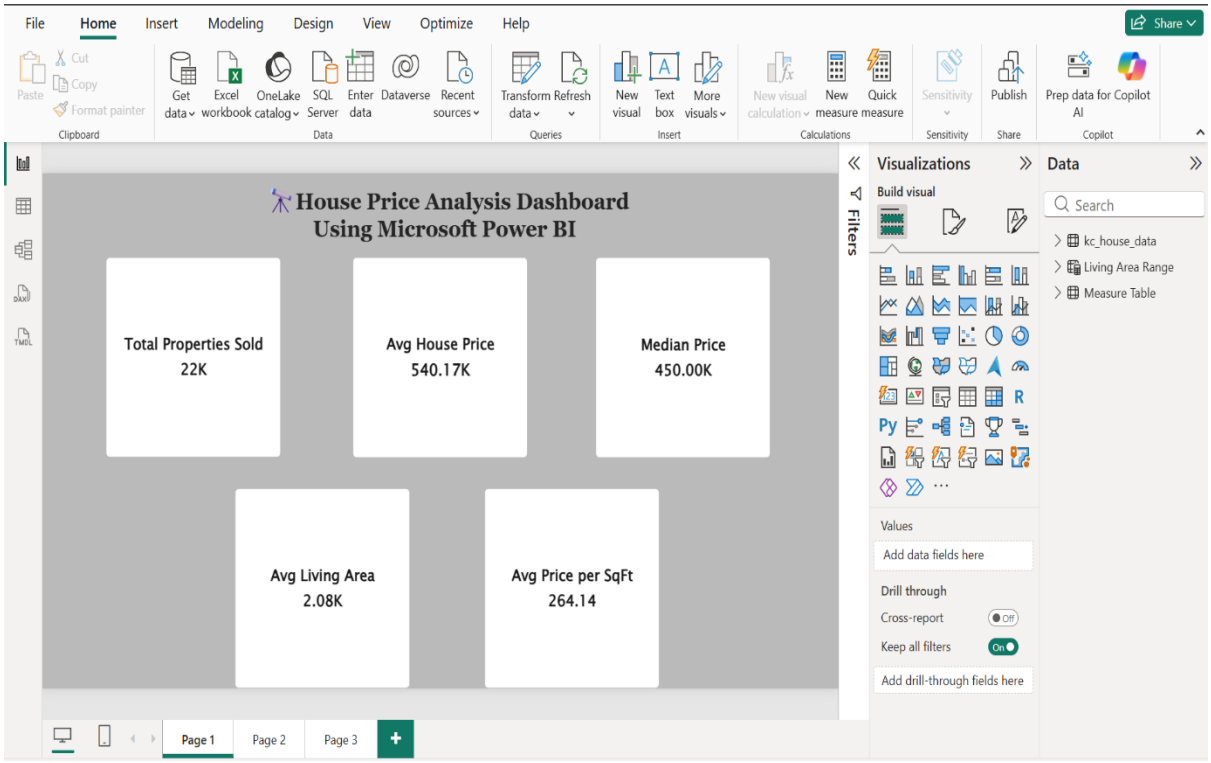
The median price card displays the middle value of house prices. It helps understand the typical property price without being heavily affected by extremely high or low prices.

6.4 Average Living Area

This KPI displays the average living area of the properties. It provides information about the average size of houses.

6.5 Average Price per SqFt

This card displays the average price per square foot. It helps compare the price of properties based on their living area.

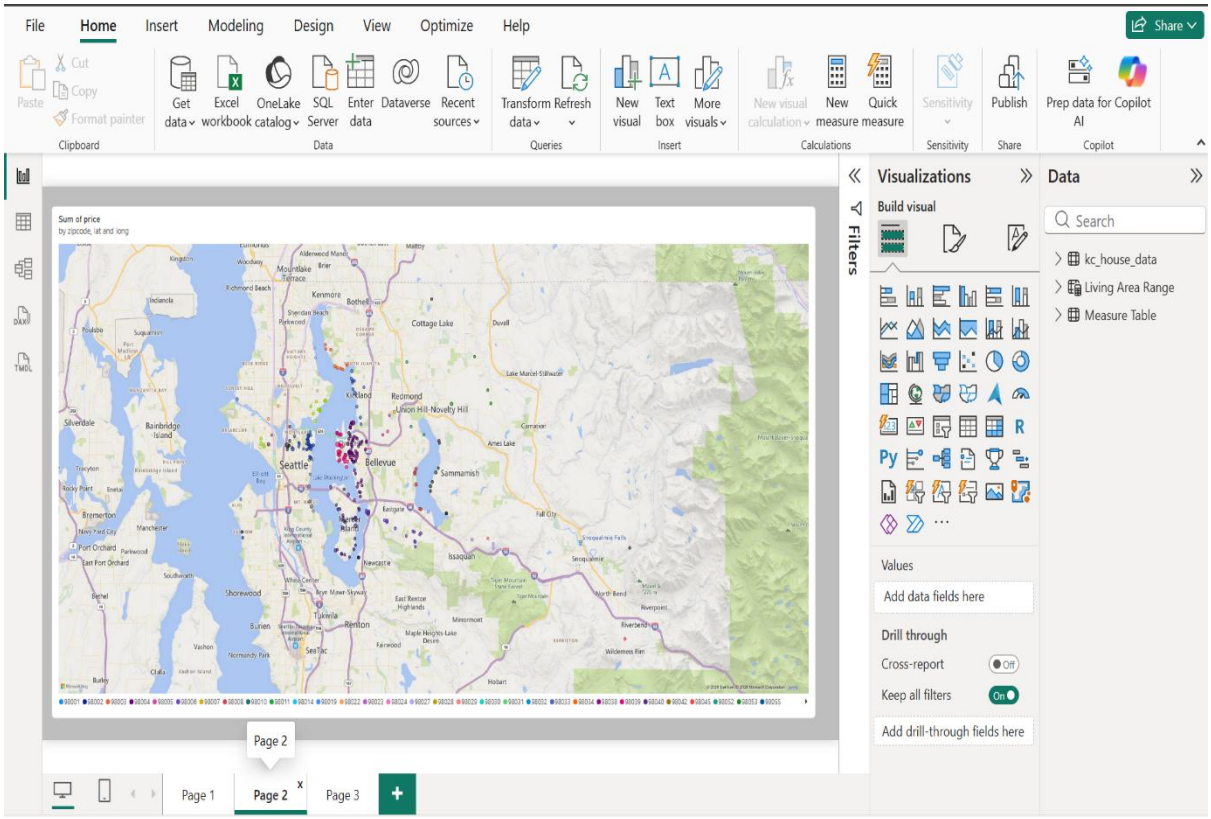


7. Page 2 – Geographical Analysis Using Map Visualizations

The second page of the dashboard focuses on geographical analysis of housing sales using two different map visualizations. These maps help to understand the location and distribution of properties across different areas.

7.1 Map Visualization – House Price by Location

The first map displays the geographical distribution of house prices using ZIP code, latitude, and longitude. Different colored points represent different ZIP code locations. The map helps identify property locations and explore house price distribution across different areas.

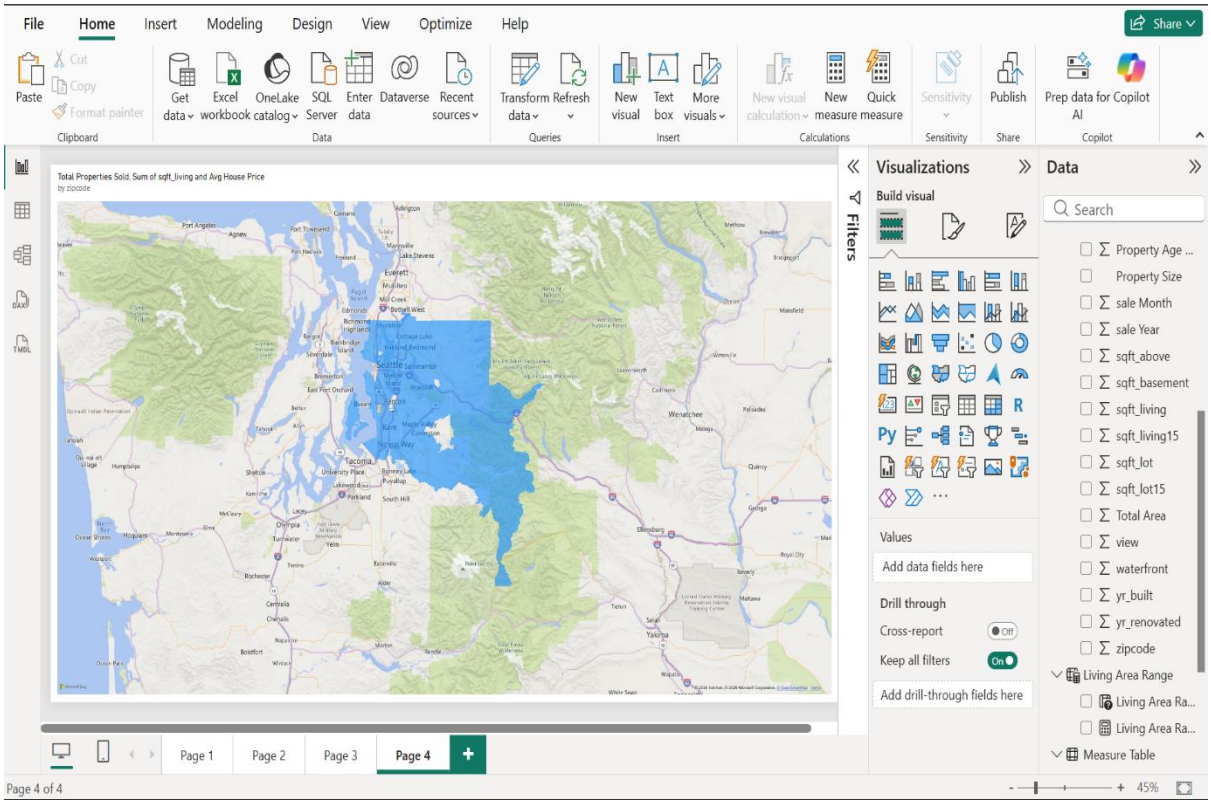


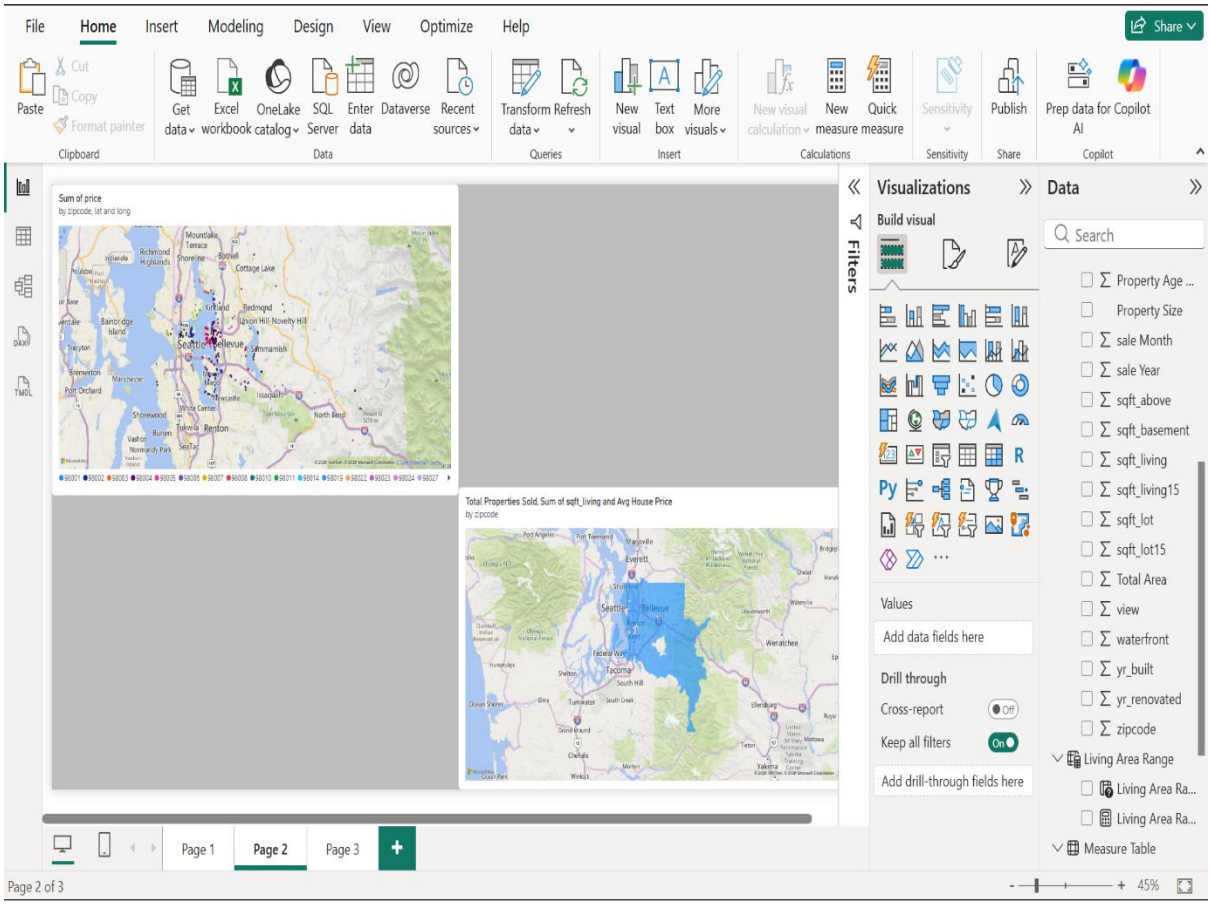
7.2 Map Visualization – Property Sales and Living Area

The second map was created using ZIP code, Total Properties Sold, Sum of SqFt Living, and Average House Price. It provides a geographical view of property sales and living area information across different locations.

This visualization helps compare housing activity across locations and understand how property sales, living area, and average house prices vary geographically.

By using two map visualizations, the dashboard provides a better understanding of the geographical distribution of housing data.


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Share
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Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
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New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
Total Properties Sold, Sum of sqft_living and Avg House Price
by zipcode
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
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Property Size
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sale Year
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sqft_basement
sqft_living
sqft_living15
sqft_lot
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Total Area
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Measure Table
Filters
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File
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Optimize
Help
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Paste
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Format painter
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Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
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New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
Sum of price
by zipcode, lat and long
Total Properties Sold, Sum of sqft_living and Avg House Price
by zipcode
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Data
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Property Size
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sale Year
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sqft_basement
sqft_living
sqft_living15
sqft_lot
sqft_lot15
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Measure Table
Filters
Page 1
Page 2
Page 3
Page 2 of 3
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8. Page 3 – Detailed Data Analysis

The third page contains multiple charts and an interactive slicer to explore different aspects of the housing dataset.

8.1 Waterfront Analysis

A chart was created to analyze waterfront properties based on house prices. It helps compare properties with different waterfront values.

8.2 Bedrooms Analysis

A visualization was created to study the relationship between the number of bedrooms and house prices. It helps understand how bedroom count is related to property prices.

8.3 Bathrooms Analysis

A chart was used to analyze house prices based on the number of bathrooms. It helps compare properties with different bathroom counts.

8.4 House Price Analysis by Bedrooms

A chart displaying average house prices based on bedrooms was created to understand price differences between properties.

8.5 House Price and Condition Analysis

A combination chart was created using house price, property ID, bedrooms, and condition. It helps explore the relationship between property condition and housing data.

8.6 Average Living Area by Bedrooms and Condition

This chart compares average living area based on the number of bedrooms and property condition.

8.7 Living Area and Total Area Analysis

A chart was created to compare average living area and total area based on house prices. It helps understand the relationship between property size and price.

8.8 Average House Price by Grade

A column chart was created to compare average house prices across different property grades. It helps understand how property grade is associated with house prices.

8.9 Average Living Area by Condition

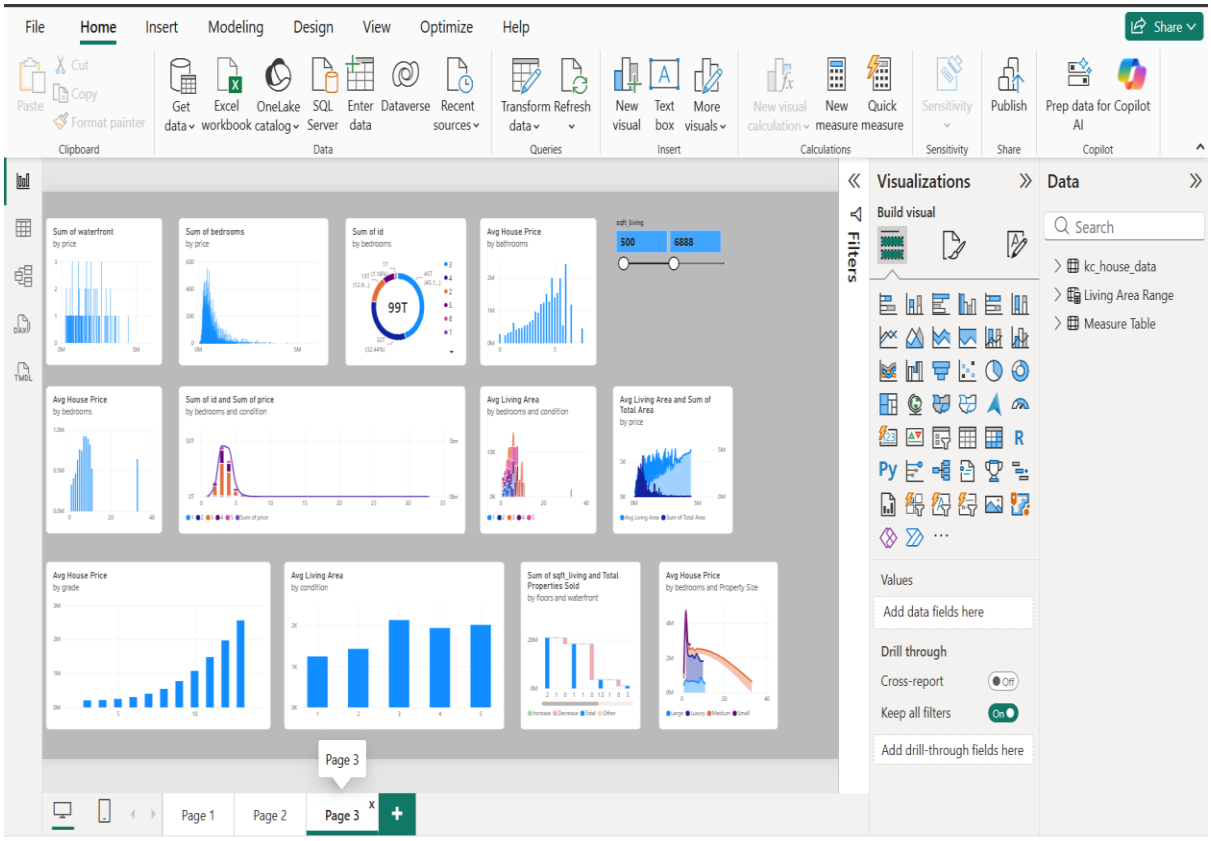
This chart displays the average living area based on property condition. It helps compare the sizes of houses across different condition categories.

8.10 Floor and Waterfront Analysis

A chart was created using floors, waterfront, and total properties sold to explore property distribution based on these features.

8.11 House Price by Bedrooms and Property Size

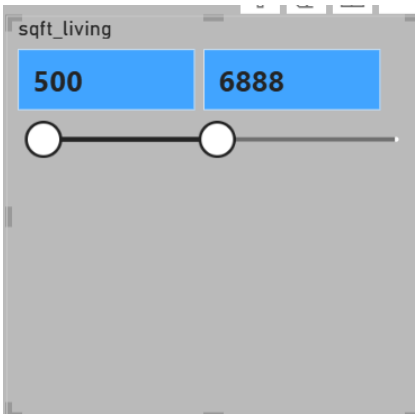
A visualization was created to compare average house prices based on bedroom count and property size categories such as Large, Luxury, Medium, and Small.



9. Slicer and Interactive Filtering

A Living Area slicer was added to the dashboard using the sqft_living field. It allows users to select a required living area range.

The selected range can be used to filter the visualizations and analyze properties according to their living area



10. DAX Measures

A separate Measure Table was created to organize the important DAX measures used in the dashboard.

The following measures were created:

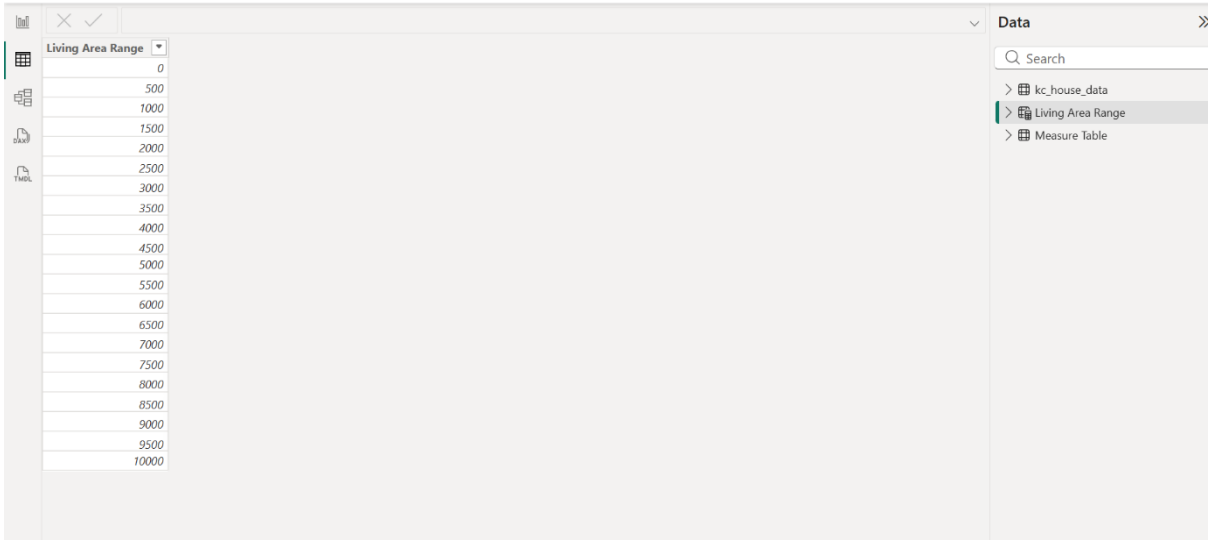
- Total Properties Sold
- Average House Price
- Median Price
- Average Living Area
- Average Price per SqFt
- Living Area Filter

These measures help calculate important values and display them in KPI cards and other visualizations.

11. Living Area Range Table

A separate table named Living Area Range was created with values ranging from 0 to 10,000 at intervals of 500.

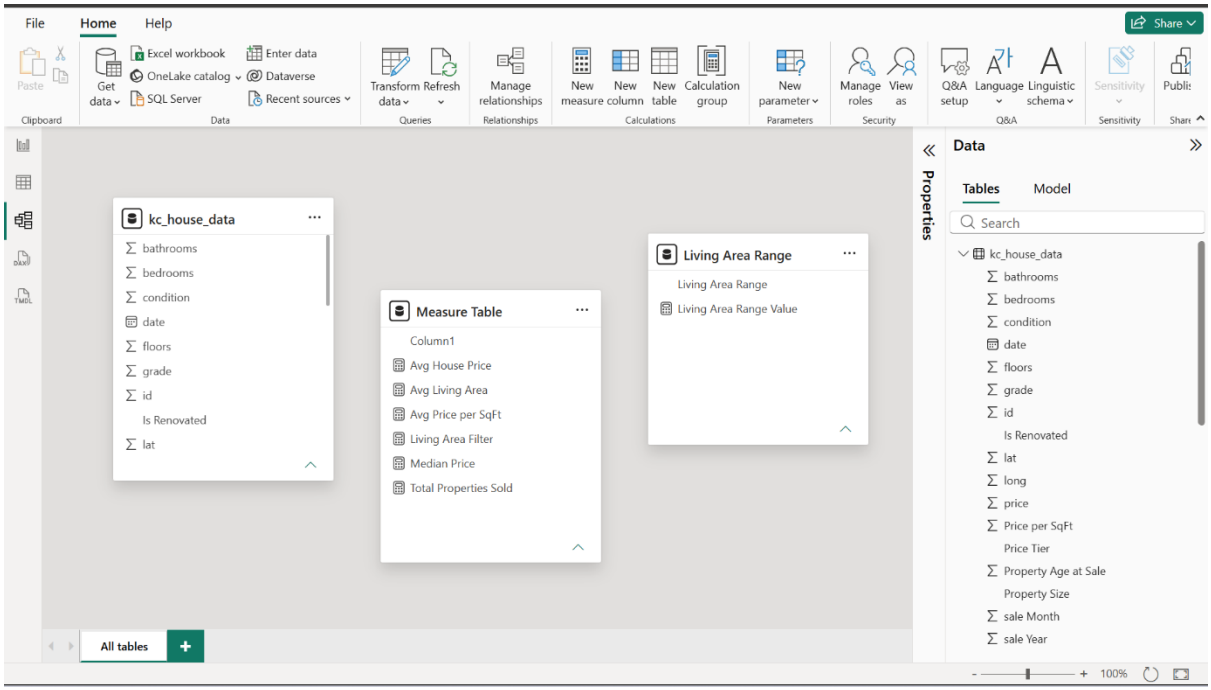
This table supports the Living Area selection and helps users filter properties based on their preferred living area range.



12. Data Model View

The Model View was used to display the tables created for the project. It includes the main kc_house_data table, Measure Table, and Living Area Range table.

The Measure Table stores the DAX measures, while the Living Area Range table supports the living area selection feature.



13. Results and Discussion

The House Price Analysis Dashboard was successfully created using Power BI. The dashboard provides information about total properties sold, average house price, median price, average living area, and average price per square foot.

The geographical map helps explore property locations, while the charts provide comparisons based on bedrooms, bathrooms, condition, grade, floors, waterfront, and property size.

The Living Area slicer makes the dashboard interactive and allows users to explore data based on selected living area values.

14. Conclusion

This project helped me gain practical knowledge of creating interactive dashboards using Microsoft Power BI. I learned how to create KPI cards, different charts, map visualizations, slicers, DAX measures, and calculated tables.

The project also improved my understanding of analyzing housing sales data and presenting meaningful insights through visualizations. Overall, the dashboard provides a clear and interactive way to explore house price information and property characteristics.